Definitions of Reciprocal Trigonometric Functions

Exam code: 9MA0

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Definitions of reciprocal trigonometric functions

What are the reciprocal trigonometric functions?

- There are three reciprocal trig functions corresponding to the three regular trig functions

1. SECANT (sec)

$$\sec x = \frac{1}{\cos x}$$

(UNDEFINED FOR VALUES OF x FOR WHICH $\cos x = 0$)

2. COSECANT (cosec)

$$\operatorname{cosec} x = \frac{1}{\sin x}$$

(UNDEFINED FOR VALUES OF x FOR WHICH $\sin x = 0$)

3. COTANGENT (cot)

$$\cot x = \frac{1}{\tan x}$$

(UNDEFINED FOR VALUES OF x FOR WHICH $\tan x = 0$; EQUAL TO ZERO FOR VALUES OF x FOR WHICH $\tan x$ IS UNDEFINED)

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Got it

$\cot x = \frac{\quad}{\sin x}$

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Examiner Tips and Tricks

- To solve equations with the reciprocal trig functions, convert them into the regular trig functions and solve in the usual way.
- Don't forget that both **tan** and **cot** can be written in terms of **sin** and **cos** (see the Worked Example below).
- You will sometimes see **csc** instead of **cosec** for cosecant.

Worked Example

a) Prove that $\frac{1+\cot x}{1+\tan x} \equiv \cot x$.

b) Hence solve, in the interval $0 \le x \le 360°$, the equation $\frac{1+\cot x}{1+\tan x} = 10$.

Give your answers to one decimal place.

a) $1+\cot x \equiv 1+\frac{\cos x}{\sin x} \equiv \frac{\sin x+\cos x}{\sin x}$

$1+\tan x \equiv 1+\frac{\sin x}{\cos x} \equiv \frac{\cos x+\sin x}{\cos x}$

REWRITE IN TERMS OF sin AND cos

SO $\frac{1+\cot x}{1+\tan x} \equiv \frac{\left(\frac{\sin x+\cos x}{\sin x}\right)}{\left(\frac{\cos x+\sin x}{\cos x}\right)} \equiv \frac{\cos x}{\sin x} \equiv \cot x$

$\frac{a/b}{a/c} = \frac{c}{b}$

b) WE NEED TO SOLVE $\cot x = 10$

$\frac{1}{\tan x} = 10$

$\tan x = \frac{1}{10}$

$x = 5.71°$

USE $\tan^{-1}$ ON YOUR CALCULATOR TO FIND THE PRINCIPAL VALUE OF x

OR $x = 180 + 5.71 = 185.71°$

THEN BE SURE TO FIND ANY OTHER VALUES IN THE REQUIRED INTERVAL

SO $x = 5.7°$ OR $185.7°$ (1 d.p.)

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